

Abhinav Kataria

Frontend Developer | React & Real-Time Interfaces

India | rishu2300@gmail.com | +91 9958390252 | github.com/realcyclopse | linkedin.com/in/realcyclopse | Portfolio | LeetCode

SUMMARY

Final-year Computer Science student who builds real-time React interfaces. Scaffolded and led the **frontend architecture** of NexRoute, a live delivery-allocation platform - routing, theming, and a **Mapbox GL live map** streaming rider positions over WebSockets - across **31 of 108** commits. Also built a production-style FastAPI backend engine at Jakson Group as **sole engineer**, giving direct experience designing the REST APIs a frontend consumes. Strong **C++ DSA** base (500+ problems, rank **1769**, LeetCode Biweekly 180).

EDUCATION

Birla Institute of Technology, Mesra

2023 - 2027

B.Tech in Computer Science Engineering | **Coursework:** DSA, OOP, DBMS, Operating Systems, Computer Networks

Vishwa Bharati Public School, Noida

2008 - 2022

CBSE Class XII: **86.2%** | Class X: **95%**

TECHNICAL SKILLS

Frontend: React 18, Next.js, JavaScript (ES6+), Vite, React Router, Material UI (MUI) theming, Tailwind CSS, CSS Modules, HTML5/CSS3

Real-Time & Maps: Mapbox GL (react-map-gl), Socket.IO (client), Axios/REST integration

Backend & Data (API context): Node.js, Express, FastAPI, MongoDB/Mongoose, SQL, JWT/OAuth 2.0, Role-Based Access Control, API hardening (Helmet, CORS, rate limiting, mongo-sanitize)

Languages: Python, C++, Java, C, JavaScript, SQL

Tooling: Git/GitHub, CI/CD, Vercel/Render, pytest

WORK EXPERIENCE

IT Intern, Jakson Limited

June 2026 - July 2026

Platform-Agnostic Enterprise AI Chatbot Engine - sole engineer

- Designed a **nine-stage synchronous request pipeline** (FastAPI + Uvicorn) exposing deterministic, authorization-gated REST endpoints - the same class of API a frontend team integrates against.
- Architected a single neutral engine serving **multiple internal platforms** (LMS, HRMS) from one codebase using **8 JSON config files** per platform, with zero engine-code changes to onboard a new one.
- Built a **two-point, fail-closed** PII minimization layer - inbound redaction plus outbound allowlist field-projection - so no raw record reaches the external LLM provider.
- Implemented hybrid intent resolution over **local sentence-transformers embeddings** with confidence/margin gating, routing ambiguous queries into a bounded disambiguation loop instead of guessing.
- Cut avoidable LLM calls by routing FAQ/navigation queries through a deterministic knowledge path backed by **relational (SQL) lookups via pyodbc**.
- Validated the system with **69 automated pytest tests across 14 modules (100% passing)**.

PROJECTS

NexRoute - Real-Time Delivery Partner Allocation Engine

Live Demo | GitHub

- Scaffolded the **React frontend** from scratch (Vite + React Router with protected routes) - app shell, layouts, sidebar/topbar navigation, and six core pages (Dashboard, Live Map, Orders, Riders, Settings, Allocation History).
- Built the **live map experience** with **Mapbox GL** (react-map-gl) rendering real-time rider positions and delivery routes streamed over **Socket.IO**, and fixed a bug that kept drawing routes for orders whose rider had already been reallocated.
- Designed the app's **theming system** (MUI theme config + CSS design tokens) and used it to restyle the dashboard, sidebar navigation, and map panels for visual consistency.
- Hardened the frontend: removed a **hardcoded Mapbox access token** in favor of environment config across all map pages, and wired **JWT auth** into the Socket.IO client connection alongside REST requests.
- Fixed a manual-allocation correctness bug by filtering candidates to **IDLE-only riders** before scoring - mirroring the guard the automatic tick already applied, so a busy rider can no longer be re-scored for a new order.
- Hardened the Express backend with **Helmet, CORS allowlisting, rate limiting, and mongo-sanitize** middleware, plus fail-fast startup validation for required environment variables.
- Authored **31 of 108 commits**, including a **runtime orphan-order sweeper** on the backend that re-queues orders left stranded by a crashed or partial write instead of letting them stall or wrongly cancel.

POSITION OF RESPONSIBILITY & ACHIEVEMENTS

- Technical Lead, IEEE Student Chapter** and **Technical Member, GDSC**, BIT Mesra
- Technical:** Solved **500+ DSA problems in C++**; rank **1769** in LeetCode Biweekly Contest 180 (all problems solved).
- Academic:** Qualified JEE Main & Advanced 2023 (top 4%, category); cleared NTSE Level 1 (2019).